DESIGN SYSTEMS • ENTERPRISE SAAS • PRODUCT UI

Designing Decisions That Scale

Building a scalable design system for enterprise SaaS — a system for making product decisions consistently, intentionally, and at scale.

Design System
Aa
Typography
Components
Enterprise Dashboard
This Month
Revenue
$24.8M
↑ 12.5%
Users
12,845
↑ 8.2%
Conversion
3.42%
↑ 16.7%
Active Now
256
↑ 5.4%
Performance Overview
Top Products
Product A 48%
Product B 28%
Product C 16%
Others 8%

ROLE
UI/UX Designer
Product Designer

FOCUS
Design Systems
Enterprise SaaS
Product UI

TOOLS
Figma
Prototyping
Design System

Design once. Define the rule. Scale with confidence.

A system for making product decisions consistently, intentionally, and at scale.

01 CONTEXT & CHALLENGE

The Problem We Were Solving

As enterprise SaaS products grow, visual inconsistency rarely comes from one major mistake. It accumulates through hundreds of small decisions.

Seven Core Problems

- Inconsistent UI patterns across different product areas making the experience feel fragmented
- Repeated design decisions — the same problems solved differently each time by different teams
- No shared design language between designers and developers
- Typography and spacing applied arbitrarily, not systematically
- Color usage without semantic meaning or consistent application
- Components built from scratch each sprint rather than reused from a shared library
- No scalable foundation — every new feature required rediscovering basic design rules

Individually, these decisions may seem insignificant. At scale, they create a product that becomes increasingly difficult to maintain and evolve.

The Challenge

The goal was to establish a shared design foundation that could support an enterprise SaaS product without turning the system into an overly rigid collection of rules. The system needed to make everyday design decisions: more consistent, more predictable, and easier to scale.

The Design Question

Instead of asking *"How should this screen look?"* — I reframed the problem as *"What rules should determine how future screens are designed?"* That shift became the foundation of the entire project.

Objective

Create a scalable design system that:

- ✓ Establishes a consistent visual language
- ✓ Reduces repeated design decisions
- ✓ Supports dense enterprise interfaces
- ✓ Provides clear rules without restricting creativity
- ✓ Translates foundations into practical product decisions

CORE PRINCIPLE
Design the decision once. Make it reusable everywhere

02 DESIGN APPROACH

How We Approached the System

The system was approached as a hierarchy of decisions rather than a collection of isolated UI assets. The process moved through four levels.

Phase 01
Understand
Identify recurring visual and structural decisions across the product.

Phase 02
Define
Establish principles for hierarchy, meaning, spacing, structure, shape, and depth.

Phase 03
Systemize
Translate those decisions into reusable foundations and semantic rules.

Phase 04
Apply
Test the system against product scenarios to validate the foundations worked together.

Four Principles

These principles ensured the design system is a scalable system of decisions — not just a collection of tokens and components.

01
Consistency
Repeated decisions should produce predictable outcomes.
- ✓ Same rules, same results
- ✓ Predictable across products
- ✓ Reduces visual inconsistency

02
Meaning
Design tokens should communicate intent, not simply store values.
- ✓ Colors have purpose
- ✓ Spacing conveys hierarchy
- ✓ Components communicate role

03
Flexibility
Constraints without forcing every interface into the same composition.
- ✓ Enough freedom to adapt
- ✓ Multiple layouts, one structure
- ✓ Supports different use cases

04
Scalability
New product experiences can use the same foundations without recreating the visual language.
- ✓ Built for growth
- ✓ Easy to extend
- ✓ Future-ready foundation

KEY STATEMENT
We are not designing screens. We are designing the rules that screens follow.

03 SYSTEM ARCHITECTURE

Four Layers of Decision

The system is built on a four-layer architecture that flows from raw values to product-ready decisions.

1 · FOUNDATION
Defines the raw design values—colors, typography, Spacing, sizing, Radius responsive, borders, Opacity, elevation and Grid.
"What values exist?"

→

2 · SEMANTIC
Assigns meaning to foundation values through product roles such as Primary, Secondary, Accent, and Neutral.
"What do these values represent?"

→

3 · COMPONENT
Applies semantic decisions to UI components, defining how tokens behave across states, interactions, and usage contexts.
"How should these meanings be applied?"

→

04 · Product UI
Brings the system together in real interfaces through patterns, layouts, and product experiences.
"How does it come together in the UI?"

System Architecture Insight: The architecture separates raw values, semantic meaning, component behavior, and product application into distinct layers. This keeps the system flexible and scalable changes at the foundation can flow through semantic and component layers without requiring changes to the underlying product UI.

SYSTEM ARCHITECTURE INSIGHT
Foundation define the values → Semantic tokens define the meaning →Components define the behavior →Product UI defines the context.

3.1 FOUNDATIONS -- RAW VALUES

Building the Visual Foundation

The foundation layer defines the core visual decisions used consistently across the design system.

1 Color
It contains the base color data—such as brand, neutral, semantic, and supporting color values—used to build semantic color tokens

2 Typography
Raw type definitions that establish font families, sizes, weights, line heights, and letter spacing across the system

3 Spacing
Base spacing values that define consistent gaps and distances between elements and sections.

4 Sizing
Raw size values that define consistent widths, heights, and dimensions for UI elements.

5 Responsive
Breakpoint values and responsive rules that define how layouts adapt across different screen sizes.

6 Radius
Base corner-radius values used to create consistent rounding across UI elements.

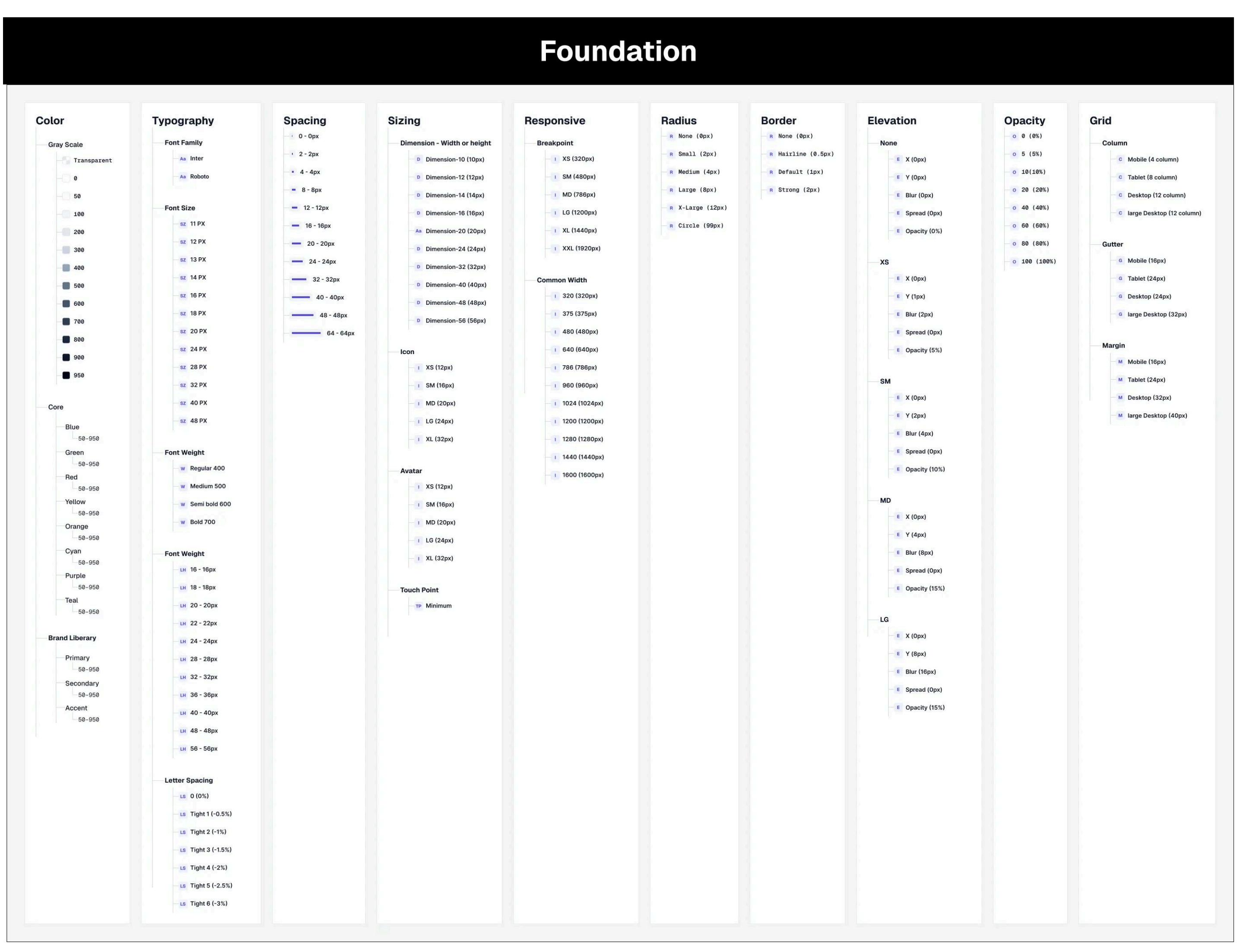
7 Border
Raw border values defining widths and styles used for containers, controls, and UI elements.

8 Elevation
Base shadow and elevation values that establish depth and hierarchy between interface layers.

9 Opacity
Raw transparency values used to control visibility, emphasis, overlays, and disabled states.

10 Grid
Base grid definitions that establish columns, gutters, margins, and layout structure for consistent page composition.

Foundation Token Architecture


Foundation
Color
Gray Scale
Transparent
0
50
100
200
300
400
500
600
700
800
900
950
Core
Blue 50-950
Green 50-950
Red 50-950
Yellow 50-950
Orange 50-950
Cyan 50-950
Purple 50-950
Teal 50-950
Brand Liberary
Primary 50-950
Secondary 50-950
Accent 50-950
Typography
Font Family
Inter
Roboto
Font Size
11 PX
12 PX
13 PX
14 PX
16 PX
18 PX
20 PX
24 PX
28 PX
32 PX
40 PX
48 PX
Font Weight
Regular 400
Medium 500
Semi bold 600
Bold 700
Font Weight
16 - 16px
18 - 18px
20 - 20px
22 - 22px
24 - 24px
28 - 28px
32 - 32px
36 - 36px
40 - 40px
48 - 48px
56 - 56px
Letter Spacing
0 (0%)
Tight 1 (-0.5%)
Tight 2 (-1%)
Tight 3 (-1.5%)
Tight 4 (-2%)
Tight 5 (-2.5%)
Tight 6 (-3%)
Spacing
0 - 0px
2 - 2px
4 - 4px
8 - 8px
12 - 12px
16 - 16px
20 - 20px
24 - 24px
32 - 32px
40 - 40px
48 - 48px
64 - 64px
Sizing
Dimension - Width or height
Dimension-10 (10px)
Dimension-12 (12px)
Dimension-14 (14px)
Dimension-16 (16px)
Dimension-20 (20px)
Dimension-24 (24px)
Dimension-32 (32px)
Dimension-40 (40px)
Dimension-48 (48px)
Dimension-56 (56px)
Icon
XS (12px)
SM (16px)
MD (20px)
LG (24px)
XL (32px)
Avatar
XS (12px)
SM (16px)
MD (20px)
LG (24px)
XL (32px)
Touch Point
Minimum
Responsive
Breakpoint
XS (320px)
SM (480px)
MD (786px)
LG (1200px)
XL (1440px)
XXL (1920px)
Common Width
320 (320px)
375 (375px)
480 (480px)
640 (640px)
786 (786px)
960 (960px)
1024 (1024px)
1200 (1200px)
1280 (1280px)
1440 (1440px)
1600 (1600px)
Radius
None (0px)
Small (2px)
Medium (4px)
Large (8px)
X-Large (12px)
Circle (99px)
Border
None (0px)
Hairline (0.5px)
Default (1px)
Strong (2px)
Elevation
None
X (0px)
Y (0px)
Blur (0px)
Spread (0px)
Opacity (0%)
XS
X (0px)
Y (1px)
Blur (2px)
Spread (0px)
Opacity (5%)
SM
X (0px)
Y (2px)
Blur (4px)
Spread (0px)
Opacity (10%)
MD
X (0px)
Y (4px)
Blur (8px)
Spread (0px)
Opacity (15%)
LG
X (0px)
Y (8px)
Blur (16px)
Spread (0px)
Opacity (15%)
Opacity
0 (0%)
5 (5%)
10(10%)
20 (20%)
40 (40%)
60 (60%)
80 (80%)
100 (100%)
Grid
Column
Mobile (4 column)
Tablet (8 column)
Desktop (12 column)
large Desktop (12 column)
Gutter
Mobile (16px)
Tablet (24px)
Desktop (24px)
large Desktop (32px)
Margin
Mobile (16px)
Tablet (24px)
Desktop (32px)
large Desktop (40px)

3.2 SEMANTIC -- PRODUCT IDENTITY

Defining Semantic Meaning

Semantic tokens translate foundation values into purposeful roles, creating a consistent language for text, surfaces, actions, states, interactions, and data visualization.

1 Text
Defines semantic roles for text, including primary, secondary, muted, and inverse content.

2 Surface
Defines semantic roles for backgrounds, containers, cards, overlays, and interactive surfaces.

3 Border
Defines semantic roles for borders and dividers across default, subtle, strong, and interactive states.

4 Icon
Defines semantic roles for icons based on hierarchy, context, state, and interaction.

5 Action
Defines semantic roles for interactive actions such as primary, secondary, neutral, and destructive actions.

6 Link
Defines semantic roles for links across default, hover, active, visited, and disabled states.

7 Status
Defines semantic roles for feedback states such as information, success, warning, and error.

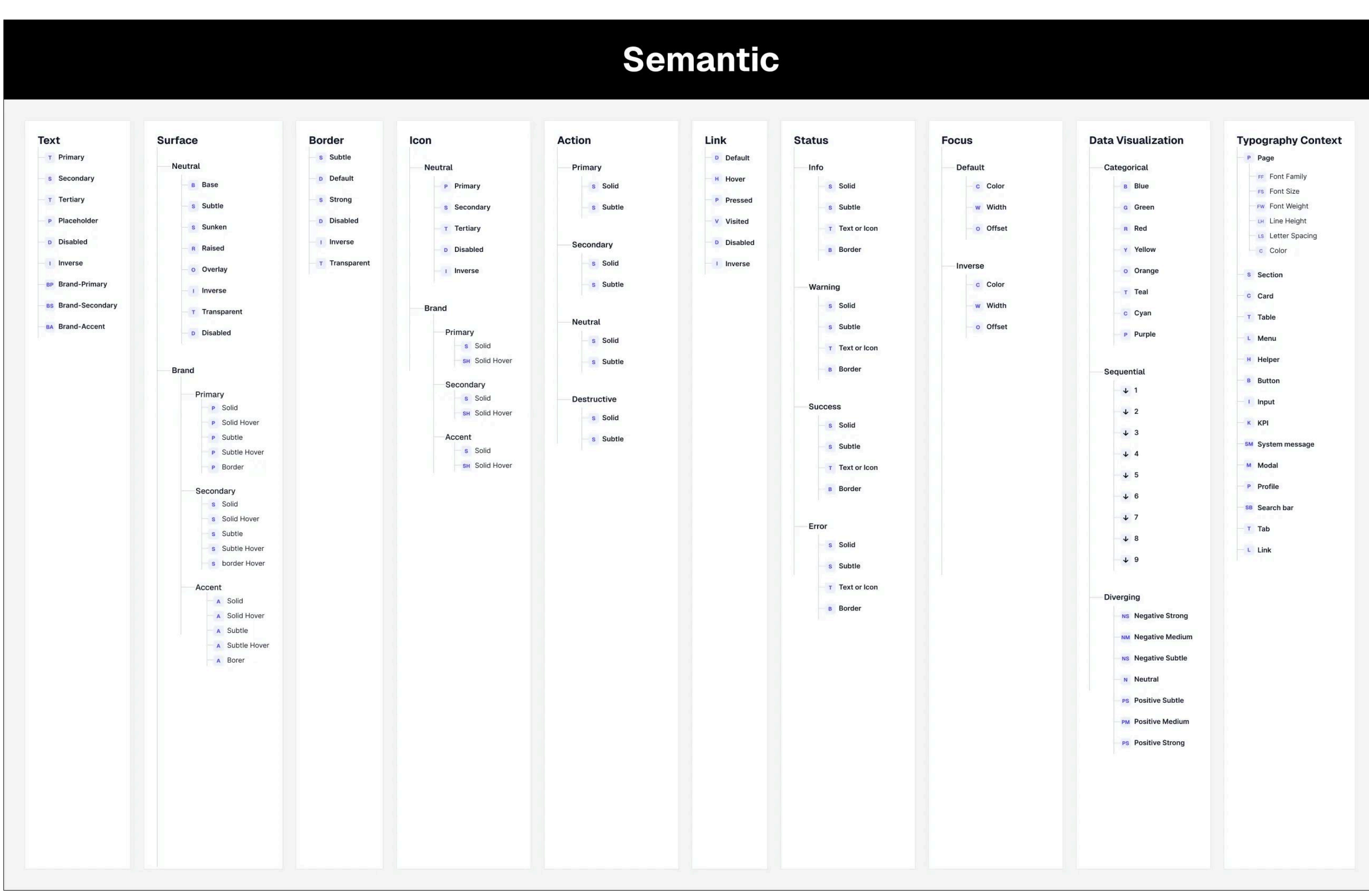
8 Selection
Defines semantic roles for selected, unselected, checked, and interactive selection states.

9 Focus
Defines semantic roles for focus indicators and keyboard interaction states.

7 Data Visualization
Defines semantic roles for categorical, sequential, and diverging data visualization colors.

8 Typography
Defines semantic text styles for headings, body content, labels, captions, and other content hierarchies.

Semantic Token Architecture


Semantic
Text
Primary
Secondary
Tertiary
Placeholder
Disabled
Inverse
Brand-Primary
Brand-Secondary
Brand-Accent
Surface
Neutral
Base
Subtle
Sunken
Raised
Overlay
Inverse
Transparent
Disabled
Brand
Primary
Solid
Solid Hover
Subtle
Subtle Hover
Border
Secondary
Solid
Solid Hover
Subtle
Subtle Hover
border Hover
Accent
Solid
Solid Hover
Subtle
Subtle Hover
Borer
Border
Subtle
Default
Strong
Disabled
Inverse
Transparent
Icon
Neutral
Primary
Secondary
Tertiary
Disabled
Inverse
Brand
Primary
Solid
Solid Hover
Secondary
Solid
Solid Hover
Accent
Solid
Solid Hover
Action
Primary
Solid
Subtle
Secondary
Solid
Subtle
Neutral
Solid
Subtle
Destructive
Solid
Subtle
Link
Default
Hover
Pressed
Visited
Disabled
Inverse
Status
Info
Solid
Subtle
Text or Icon
Border
Warning
Solid
Subtle
Text or Icon
Border
Success
Solid
Subtle
Text or Icon
Border
Error
Solid
Subtle
Text or Icon
Border
Focus
Default
Color
Width
Offset
Inverse
Color
Width
Offset
Data Visualization
Categorical
Blue
Green
Red
Yellow
Orange
Teal
Cyan
Purple
Sequential
1
2
3
4
5
6
7
8
9
Diverging
Negative Strong
Negative Medium
Negative Subtle
Neutral
Positive Subtle
Positive Medium
Positive Strong
Typography Context
Page
Font Family
Font Size
Font Weight
Line Height
Letter Spacing
Color
Section
Card
Table
Menu
Helper
Button
Input
KPI
System message
Modal
Profile
Search bar
Tab
Link

3.3 COMPONENT -- MEANING IN CONTEXT

Building Meaningful Components

The component layer translates semantic decisions into reusable UI patterns, behaviors, and interaction states.

1 Button
Provides actions for users, with clear hierarchy, interaction states, and loading or disabled behavior.

2 Input
Enables users to enter and edit information with validation, states, labels, and supporting feedback.

3 Checkbox
Allows users to select one or multiple options with checked, unchecked, indeterminate, and disabled states.

4 Radio
Allows users to select one option from a mutually exclusive set with clear selection and interaction states.

5 Badge
Communicates concise status, category, count, or contextual information within the interface.

6 Tooltip
Provides brief contextual information or guidance when users interact with or hover over an element.

7 Tab
Organizes related content into navigable sections while clearly indicating the active view.

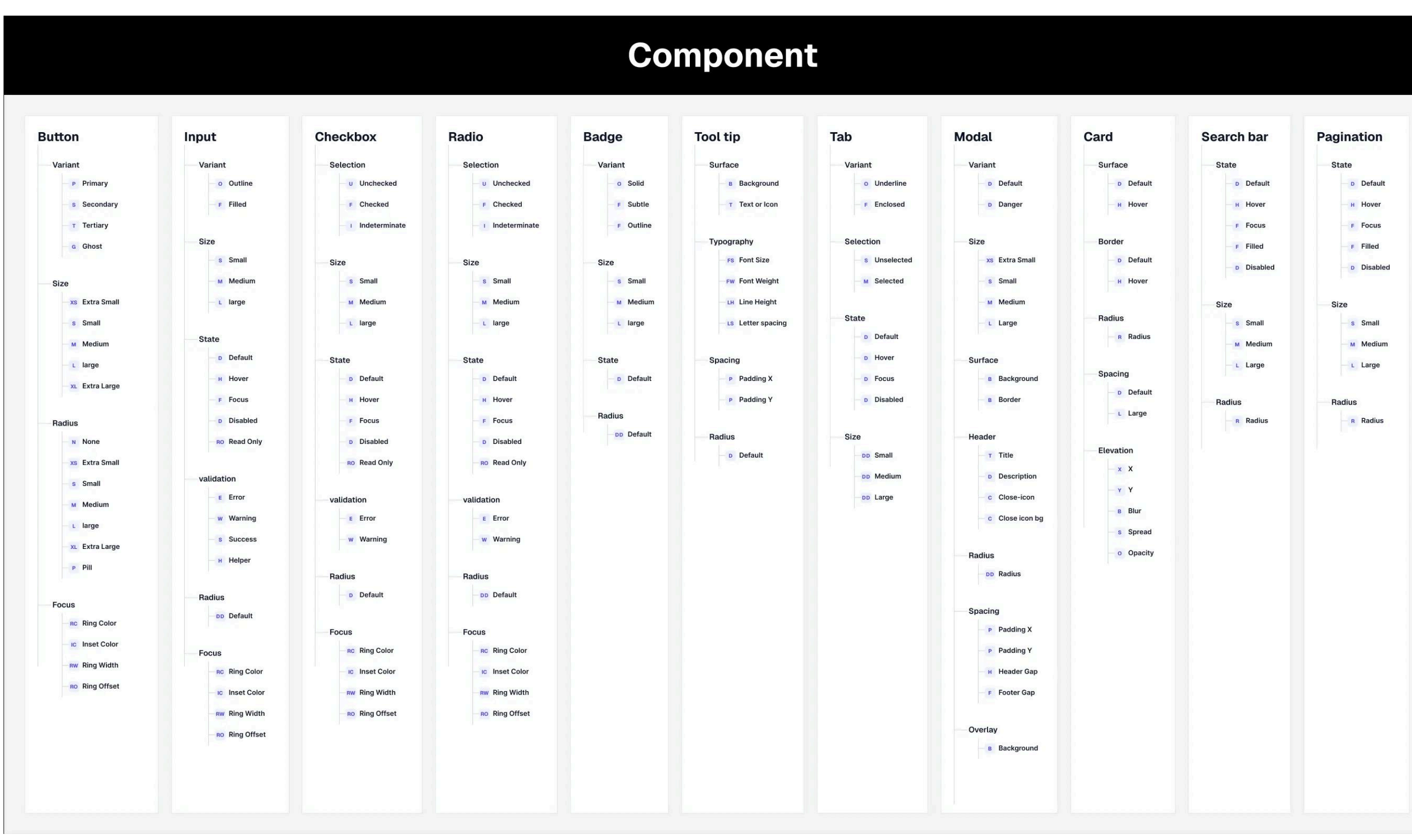
8 Modal
Presents focused content, tasks, or decisions while temporarily interrupting the underlying page context.

9 Card
Groups related content or actions into a structured container that supports hierarchy and scannability.

7 Search bar
Helps users discover content by entering queries with clear input, interaction, and result states.

8 Pagination
Enables users to navigate through large datasets or content across multiple pages.

Component Token Architecture


Component
Button
Variant
Primary
Secondary
Tertiary
Ghost
Size
Extra Small
Small
Medium
large
Extra Large
Radius
None
Extra Small
Small
Medium
large
Extra Large
Pill
Focus
Ring Color
Inset Color
Ring Width
Ring Offset
Input
Variant
Outline
Filled
Size
Small
Medium
large
State
Default
Hover
Focus
Disabled
Read Only
validation
Error
Warning
Success
Helper
Radius
Default
Focus
Ring Color
Inset Color
Ring Width
Ring Offset
Checkbox
Selection
Unchecked
Checked
Indeterminate
Size
Small
Medium
large
State
Default
Hover
Focus
Disabled
Read Only
validation
Error
Warning
Radius
Default
Focus
Ring Color
Inset Color
Ring Width
Ring Offset
Radio
Selection
Unchecked
Checked
Indeterminate
Size
Small
Medium
large
State
Default
Hover
Focus
Disabled
Read Only
validation
Error
Warning
Radius
Default
Focus
Ring Color
Inset Color
Ring Width
Ring Offset
Badge
Variant
Solid
Subtle
Outline
Size
Small
Medium
large
State
Default
Radius
Default
Tool tip
Surface
Background
Text or Icon
Typography
Font Size
Font Weight
Line Height
Letter spacing
Spacing
Padding X
Padding Y
Radius
Default
Tab
Variant
Underline
Enclosed
Selection
Unselected
Selected
State
Default
Hover
Focus
Disabled
Size
Small
Medium
Large
Modal
Variant
Default
Danger
Size
Extra Small
Small
Medium
Large
Surface
Background
Border
Header
Title
Description
Close-icon
Close icon bg
Radius
Radius
Spacing
Padding X
Padding Y
Header Gap
Footer Gap
Overlay
Background
Card
Surface
Default
Hover
Border
Default
Hover
Radius
Radius
Spacing
Default
Large
Elevation
X
Y
Blur
Spread
Opacity
Search bar
State
Default
Hover
Focus
Filled
Disabled
Size
Small
Medium
Large
Radius
Radius
Pagination
State
Default
Hover
Focus
Filled
Disabled
Size
Small
Medium
Large
Radius
Radius

04 · PRODUCT UI

Applying the System to Product UI

The product layer brings foundations, semantic decisions, and components together to create consistent, scalable, and purposeful user experiences.

1 Patterns
Reusable interaction patterns that solve recurring product workflows and user needs.

2 Layouts
Structured page compositions that establish hierarchy, spacing, alignment, and information density.

3 Navigation
Consistent navigation patterns that help users understand where they are and move through the product efficiently.

4 States & Feedback
Product-level experiences for loading, empty, success, warning, error, disabled, and other system states.

5 States & Feedback
Product-level experiences for loading, empty, success, warning, error, disabled, and other system states.

6 Data & Information
Structured presentation of complex information through tables, cards, dashboards, filters, and visualizations.

7 Responsive Experience
Product layouts and interactions adapted across screen sizes while preserving hierarchy and usability.

8 Accessibility
Product experiences designed to remain usable, understandable, and accessible across different user needs and interaction methods.

04 FROM FOUNDATIONS TO PRODUCT

Foundations → Semantic → Components→ Product UI

The system creates value when its foundations and decisions begin shaping real product experiences—hierarchy, density, layout, states, and interaction.

The goal was not to build every possible SaaS component. It was to demonstrate how the same underlying principles could guide real product decisions and composition.

KEY INSIGHT
A design system is not just a collection of components, it is a system of decisions that creates consistency.

05 IMPACT & LEARNINGS

From visual consistency to design efficiency

The value of the design system was not measured by the number of tokens or components created. It was measured by how effectively it turned repeated design decisions into clear, reusable rules.

Consistent Product Language
Typography, color, spacing, grid, radius, and elevation follow shared rules rather than being defined independently.

Faster Design Decisions
Designers reference established values instead of repeatedly deciding how a heading, card, state, or layout should behave.

Controlled Visual Complexity
The system provides flexibility while intentionally limiting arbitrary choices.

Scalable Foundation
New product experiences can be composed from the same foundations without recreating the visual language from scratch.

What I Learned

01 A Design System Is a Decision System
A design system is more than a collection of tokens and components. Its real value comes from how effectively it guides decisions.
Tokens define the boundaries. Principles define how those boundaries are used.

02 Consistency Doesn't Mean Uniformity
A scalable system shouldn't force every screen to look identical. The goal is to establish consistent rules while allowing contextual variation.

03 Fewer Rules Can Create More Scalability
A large collection of highly specific rules can become difficult to maintain. A focused foundation creates a more flexible system, centered around Typography · Color · Spacing · Grid · Radius · Elevation.

04 Foundations Need Contextual Validation
A token can look correct in isolation and behave differently in a real interface. The dashboard application helped validate whether the foundations worked together under realistic information density.

The Bigger Shift

Designing Screens
Solving individual problems
Repeating decisions

→

Designing Foundations
Defining reusable decisions
Applying them across product UI

FINAL LEARNING
The designer's role shifts from creating individual interfaces to creating the rules that make future interfaces easier to design well.

CLOSING

Designing the system behind the product

The project started with a need for greater consistency across a complex SaaS environment. It evolved into something more fundamental: a system for making product decisions consistently, intentionally, and at scale.

Typography → Hierarchy · Color → Meaning · Spacing → Rhythm · Grid → Structure · Radius → Shape · Elevation → Depth

"Together, these foundations created a shared visual language that could support future product decisions without requiring the visual system to be reinvented each time."